Exact Randomization Inference for Cross-Outcome Falsification Tests with Few Clusters

Ethan Xie*

August 21, 2026

Abstract

Applied researchers test identifying assumptions by asking whether an estimated treatment effect varies across outcome categories that differ in how susceptible they are to a suspected violation. We show that the cluster-robust Wald statistic for the joint null of no cross-category gradient is exactly a scalar multiple of a Hotelling T^2, so its chi-square calibration is wrong by an amount governed by the ratio of restrictions to clusters. At a design with 51 clusters and 13 post-treatment restrictions the nominal 5% test rejects a true null 38.9% of the time. We give three correctly sized alternatives: a label-permutation test that is exact in finite samples under category exchangeability, a cluster sign-change test, and a restricted wild cluster bootstrap. Permutation has a resolution floor of $2/C!$ that makes it unusable below five categories. In a crime-reporting application the chi-square test rejects at $p = 0.018$ and no correctly sized procedure rejects.

Keywords: randomization inference; cluster-robust inference; falsification tests; permutation tests; Hotelling's T^2; difference-in-differences.

JEL codes: C12, C15, C23.

*Charter School of Wilmington, Wilmington, DE. Email: ethanxie@udel.edu. Replication code and all output files underlying the tables are available at https://github.com/opxie1/Exact-Randomization-Inference-for-Cross-Outcome-Falsification-Tests-with-Few-Clusters; every number reported below names the file it comes from.

1 Introduction

Falsification tests built from several outcomes are common in applied work, and their inference is usually asymptotic. The design is familiar: a researcher suspects that an estimated effect is contaminated by some mechanism, observes several outcome categories that differ in exposure to that mechanism, and tests whether the estimated effects line up with exposure. Weisburd et al. (2024) run a version of this across crime categories graded by reporting propensity. The test is then a joint restriction on interaction coefficients, and the p-value comes from a chi-square reference distribution with cluster-robust standard errors.

That calibration fails in the sample sizes where these tests are run. We derive the exact null distribution of the cluster-robust Wald statistic for this problem and show it is ω_2 times a Hotelling T^2 with dimension m and $G-1$ degrees of freedom, where m is the number of restrictions, G is the number of clusters, and ω_2 is a fitting correction within 0.002 of one in our designs (Hotelling 1931). The chi-square reference ignores the estimation of the $m \times m$ meat matrix from G cluster scores. The resulting size distortion is not a small-sample curiosity. At $G = 51$ and $m = 13$ we measure a null rejection rate of 0.389 against a nominal 0.05, and at $G = 20$ the rate is 0.905.

The distortion is governed by m and G jointly, and the chi-square calibration is valid only when m grows slower than $G^{2/3}$. Along sequences with $m^{3/2}/G$ converging to a positive constant the size converges to a limit strictly above the nominal level, and when m/G converges to a positive fraction the size converges to one. This gives a design rule computable before any outcome is observed: at 36.4 effective clusters, size stays below 0.075 only for $m \leq 2$.

We give three procedures that are correctly sized. The first permutes the exposure labels across outcome categories. Under an exchangeability condition on the joint law of the category data arrays, this test has size at most α in finite samples for every N, T, G and C, with no distributional, clustering or cluster-count assumption. The condition is not free: with C categories the smallest attainable p-value for a two-sided quadratic statistic is $2/C!$, so the test cannot reject at the 5% level with fewer than five categories. The second procedure randomizes cluster-level sign changes and the third is the restricted wild cluster bootstrap of Cameron et al. (2008); both remain usable when C is small.

Two results concern power rather than size. The Wald statistic is not optimal against the alternative these tests are designed to detect. When the shape of the response across horizons is known up to scale, a directed statistic is uniformly most powerful invariant, and recalibrating both tests to deliver size α puts the ratio of minimum detectable responses at 0.393 at 36.4 effective clusters. Label randomization also carries a price, which we quantify: relative to a normal benchmark the permutation test needs 1.35 times the signal at $C = 5$ and 1.15 at $C = 8$.

Section 3 states the design and the null. Section 4 derives the exact null law of the Wald

statistic and the condition under which its chi-square calibration is valid. Section 5 gives the label-permutation test, its exactness condition, and its resolution floor. Section 6 gives the two cluster-level procedures. Section 7 derives local power, a closed-form minimum detectable effect, and the optimal statistic. Section 8 reports size and power over $G \in \{20, 30, 51, 100\}$ and $C \in \{3, 5, 8\}$. Section 9 applies the procedures to two panels. Appendix A contains the proofs.

2 Related work

Our exact null law is a statement about the cluster-robust Wald statistic with G clusters, which places it in the few-cluster inference literature. Cameron et al. (2008) propose the wild cluster bootstrap with the null imposed and document its size improvement over the cluster-robust t; MacKinnon and Webb (2017) characterize when it fails. Neither paper treats the joint m-restriction case in which m is comparable to G, which is where cross-outcome falsification tests sit. Carter et al. (2017) show that cluster heterogeneity reduces the number of clusters that the asymptotics effectively deliver, and define the effective cluster count we use in Section 4. Canay et al. (2017) give asymptotic validity for randomization tests when the invariance holds only in the limit, and we use their framework for the case where our finite-sample exchangeability condition fails. The finite-sample argument itself is the classical group-invariance construction, for which we follow Lehmann and Romano (2005).

The design that motivates the problem is the use of graded outcome categories to detect a treatment-induced change in how outcomes are recorded. Weisburd et al. (2024) run this contrast across crime categories. Hausman et al. (1998) formalize outcome misclassification that depends on treatment status in a cross-sectional discrete-response setting, and Denteh and Kédagni (2022) and Gutknecht and Liu (2024) study misclassified outcomes in difference-in-differences and changes-in-changes designs respectively. Freyaldenhoven et al. (2019) use an auxiliary series loading on a common latent factor to purge event-study coefficients, which is the closest existing device to the loading normalization our statistic requires. We take the estimand and the identification argument from that literature as given. This paper is about the reference distribution.

Falsification tests are also run as pre-trend tests, where Roth (2022) shows that conditioning on passing one distorts the sampling distribution of the effect estimate, and Rambachan and Roth (2023) replace the test with bounds on the post-treatment violation. Our results concern the calibration of a cross-outcome test rather than the consequences of conditioning on its outcome, and the two problems compound: a test with size 0.39 conditions on an event that is mostly noise.

3 Setup and the null

Units $i = 1, \dots, N$ are grouped into clusters $g = 1, \dots, G$; periods are $t = 1, \dots, T$; outcome categories are $c = 1, \dots, C$ with known exposure indices $d_1, \dots, d_C$. Treatment is absorbing with cohort $G_i \in \{1, \dots, T\} \cup \{\infty\}$ and $D_{it} = \mathbf{1}\{t \geq G_i\}$.

Assumption 1 (category-balanced design)**.** There is a common cell set $\mathcal{C} \subseteq \{1, \dots, N\} \times \{1, \dots, T\}$ with $|\mathcal{C}| = n_0$ on which every category is observed, and the treatment path D_{it}, the cohort G_i and the cluster map $i \mapsto g(i)$ do not depend on c.

Assumption 1 rules out designs in which categories are measured on different subsamples or at different frequencies. It is verifiable from the data layout. Section 9 reports one application where it fails, and the exact procedures below do not apply there.

Let L collect the included event-time horizons with $|L| = q$, omitting $\ell = -1$ and binning endpoints, let $X \in \mathbb{R}^{n_0 \times q}$ hold the corresponding indicators, and let $P \subseteq \{1, \dots, q\}$ with $|P| = m$ be the post-treatment block. With y_{itc} the outcome and $\delta_c = d_c - \bar{d}$ the centered exposure index, the pooled gradient regression is

$$y_{itc} = a_{ic} + p_{tc} + \sum_{\ell \in L} \beta_\ell X_{(it),\ell} + \sum_{\ell \in L} \gamma_\ell \delta_c X_{(it),\ell} + u_{itc}, \tag{3.1}$$

estimated by OLS with the CR1 cluster-robust covariance clustered on g. The hypothesis under test throughout is the joint null

$$H_0 : \gamma_P = 0, \tag{3.2}$$

that the post-treatment coefficients do not vary with exposure. This is a weak null on a parameter vector, not a sharp null of no effect for every unit, and the distinction matters for the randomization arguments in Sections 5 and 6: we state separately for each procedure which invariance delivers validity and under what condition.

The cross-category structure enters (3.1) as a single interaction contrast. The procedure tests one joint hypothesis, not C separate hypotheses, so no multiplicity adjustment arises and the error rate controlled is the per-comparison rate for (3.2).

Write M for the two-way within projection on $\mathcal{C}$, $\tilde{X} = MX$, $S = \tilde{X}'\tilde{X}$, and $b_c = S^{-1}\tilde{X}'My_c$ for the within-category event-study coefficient vector.

Lemma 1 (separability)**.** *Under Assumption 1 and* $\operatorname{rank}(\tilde{X}) = q$*, the OLS estimators in* (3.1) *satisfy* $\hat{\beta} = C^{-1}\sum_c b_c$ *and* $\hat{\gamma} = \sum_c \delta_c b_c / \sum_c \delta_c^2$.

The gradient estimator is a fixed linear combination of the per-category event studies, which is what makes the category-level randomization of Section 5 available.

The b_c are within-category event studies estimated by OLS, which under staggered adoption with heterogeneous effects need not recover a convex combination of cohort-specific effects (de Chaisemartin and D'Haultfœuille 2020; Sun and Abraham 2021). Replacing each b_c by a heterogeneity-robust estimator (Callaway and Sant'Anna 2021; Borusyak et al. 2024), or by another of the alternatives surveyed in Roth et al. (2023), leaves the randomization arguments of Sections 5 and 6 intact, because those require only that the category-level statistic be order-invariant in the sense of Theorem 1. The exact null law of Section 4 is specific to the CR1 Wald statistic from (3.1) and would have to be re-derived for another estimator.

Failing to reject (3.2) does not establish that the identifying assumption holds. Section 7 gives the minimum detectable response for each design, and the applications in Section 9 report it alongside every non-rejection.

4 The exact null law of the cluster-robust Wald statistic

The size distortion has a closed form. We isolate it under an idealization that is used only for this diagnosis and is not maintained elsewhere in the paper.

Assumption 2 (ideal cluster experiment)**.** (i) Under H_0 in (3.2), the cluster score contributions $\eta_1, \dots, \eta_G$ are i.i.d. $N(0, \Sigma)$ with $\Sigma_{PP} \succ 0$. (ii) Every cluster carries the same design weight, $S_g = S/G$ for all g, where $S_g = \tilde{X}_g' \tilde{X}_g$.

Part (i) holds when the errors in (3.1) are Gaussian, independent across clusters, and identically distributed across clusters given the design. Part (ii) fails in a staggered design with a large never-treated share; Definition 1 handles that case.

Proposition 1 (exact null law)**.** *Let Assumptions 1 and 2 hold, let* $\operatorname{rank}(\tilde{X}) = q$*, let* H_0 *in* (3.2) *hold, and let* $G - m \geq 1$*. Put* $\omega_2 = (n - 2q)/(n - 1)$ *with* $n = Cn_0$*, let* $\hat{S}_P$ *be the sample covariance of the* η_{gP} *and* $T^2 = G\bar{\eta}_P' \hat{S}_P^{-1} \bar{\eta}_P$*. Then the CR1 Wald statistic* W *for* (3.2) *satisfies* $W = \omega_2 T^2$ *identically, and*

$$\frac{(G - m)\, W}{\omega_2 (G - 1) m} \sim F(m, G - m) \tag{4.1}$$

exactly for every G *and* m *with* $G - m \geq 1$*.*

The chi-square calibration treats the estimated meat matrix as known. Under Assumption 2 the cost of that substitution is exactly the F denominator in (4.1), which has $G - m$ degrees of freedom. The statement is a degrees-of-freedom result: nothing about the errors beyond Assumption 2(i) enters, and ω_2 equals 0.9981 at the design of Section 8. Proposition 1 also supplies a valid reference distribution, since (4.1) can be inverted for a critical value.

Corollary 1 (when the chi-square calibration is valid)**.** *Let* $\operatorname{size}(G, m)$ *denote the null rejection probability of the nominal level-*α *chi-square test under the conditions of Proposition 1, and assume* $\sqrt{m}\,|\log \omega_2| \to 0$*.*

(i) For fixed m, $\text{size}(G,m) = \alpha + c_m f_m(c_m)(m+c_m)/\{2(G-1)\} + O(G^{-2})$ *where* $c_m = \chi^2_{m,1-\alpha}$ *and* f_m *is the* χ^2_m *density, so* $\text{size}(G,m) \to \alpha$ *as* $G \to \infty$.
(ii) Let $m \to \infty$ *with* $m/G \to 0$ *and* $\chi_{m,G} = \sqrt{m/2}\,\log\{(G-1)/(G-m)\} \to \chi \in [0,\infty]$. *Then* $\text{size}(G,m) \to \Phi(\chi - z_{1-\alpha})$, *and* $\chi_{m,G} = m^{3/2}/(\sqrt{2}\,G) + o(1)$ *whenever* $m^{5/2}/G^2 \to 0$. *If instead* $m/G \to r \in (0,1)$, *then* $\text{size}(G,m) \to 1$.

Part (ii) is the operative statement. The chi-square calibration is asymptotically valid if and only if $m^{3/2}/G \to 0$, that is $m = o(G^{2/3})$. The number of restrictions and the number of clusters cannot be considered separately.

Definition 1 (effective clusters)**.** For a contrast $w \in \mathbb{R}^m$ put $a = S^{-1}[\cdot, P]w$ and $\varphi_g = a' S_g a \geq 0$. Then $G_{\text{eff}}(w) = (\sum_g \varphi_g)^2 / \sum_g \varphi_g^2$.

Definition 1 is the effective cluster count of Carter et al. (2017), specialized to the contrast w. It satisfies $1 \leq G_{\text{eff}}(w) \leq G$, with equality on the right when the φ_g are equal, which is what Assumption 2(ii) imposes. Substituting G_{eff} for G in (4.1) is an approximation; Proposition 1 does not license it. Section 8 reports its accuracy. Both G_{eff} and m are computable from the design alone, before any outcome is observed, which turns Corollary 1 into an ex-ante restriction on how many horizons a researcher may test.

5 The label-permutation test

Write $\mathcal{Y}_c \in \mathbb{R}^{n_0}$ for the category-c data array, $\mathcal{D} = (\mathcal{Y}_1, \ldots, \mathcal{Y}_C)$, and $\mathcal{Z}$ for the design, which is common to all categories under Assumption 1. Let $\mathcal{T}(\mathcal{D}, \delta)$ be a statistic with large values evidence against H_0 in (3.2).

Definition 2 (label-permutation test)**.** For $\pi \in S_C$ let $\delta_\pi = (\delta_{\pi(1)}, \ldots, \delta_{\pi(C)})$ and $T(\pi) = \mathcal{T}(\mathcal{D}, \delta_\pi)$. With $M = C!$ the test rejects at level α when $\hat{p} = M^{-1}\#\{\pi \in S_C : T(\pi) \geq T(\text{id})\} \leq \alpha$.

The randomization is over the $C!$ assignments of the exposure labels to the category data arrays, holding the arrays and the design fixed. Nothing is permuted at the unit, cluster or period level.

Assumption 3 (category exchangeability)**.** Conditional on $\mathcal{Z}$, $(\mathcal{Y}_{\pi(1)}, \ldots, \mathcal{Y}_{\pi(C)}) \overset{d}{=} (\mathcal{Y}_1, \ldots, \mathcal{Y}_C)$ for every $\pi \in S_C$, and $d_1, \ldots, d_C$ are fixed constants.

Assumption 3 rules out categories with different error variances, different loadings on a common factor unless the outcome is normalized, and any dependence of the labels on the data. It is not implied by H_0 in (3.2) and it is not directly testable, though its leading failure mode has an observable signature: Section 8 prices the size distortion as a function of the ratio of the largest to smallest category error variance.

Theorem 1 (exact finite-sample validity)**.** *Suppose Assumption 3 holds and $\mathcal{T}$ is order-invariant, in the sense that it is a function of the multiset $\{(\delta_c, \mathcal{Y}_c)\}_{c=1}^{C}$ and not of the order in which the pairs are listed. Then for every N, T, G and C, the test of Definition 2 for the null* (3.2) *under the randomization over S_C satisfies* $\Pr(\textit{reject}) \leq \alpha$*, with* $\Pr(\textit{reject}) = \lfloor \alpha M \rfloor / M$ *when the M values are almost surely distinct. The randomized version has size exactly α.*

Exactness here is finite-sample exactness for the null (3.2) under the randomization distribution of Definition 2, and it requires Assumption 3. No condition on G, on the error distribution, or on the within-cluster dependence structure is used. Order-invariance holds for the OLS estimates and CR1 covariance in (3.1), and it fails for any loading normalization that selects a reference category by its position in an ordering of d.

Proposition 2 (attainable p-values)**.** *(i) $\hat{p} \geq 1/C!$ always. (ii) If $\mathcal{T}$ is even in the labels, so that $\mathcal{T}(\mathcal{D}, -\delta) = \mathcal{T}(\mathcal{D}, \delta)$, and the multiset $\{\delta_c\}$ is symmetric about zero, then $\hat{p} \geq 2/C!$. (iii) A level-α label-permutation test therefore requires $C! \geq 2/\alpha$ for the two-sided Wald statistic with equally spaced d, that is $C \geq 5$ at $\alpha = 0.05$, and $C! \geq 1/\alpha$ for a one-sided statistic, that is $C \geq 4$.*

The Wald statistic is even in the labels because $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ and its covariance estimator is quadratic in the cluster scores. Proposition 2 is a property of the randomization group, not of any implementation. With $C = 3$ the two-sided Wald has three attainable p-values, the smallest being $1/3$, and the test can never reject at conventional levels. Section 8 reports the resulting rejection rate of exactly 0.000.

6 Cluster-level randomization and the bootstrap

Two procedures remain available when $C \leq 4$. Both randomize over clusters rather than category labels, so Proposition 2 does not bind.

The first randomizes sign changes over $\mathcal{E} = \{-1, +1\}^G$, applied to the cluster contributions $\zeta_1, \ldots, \zeta_G$ to the gradient score with the null (3.2) imposed.

Theorem 2 (cluster sign-change test)**.** *If, under H_0 in* (3.2)*, the cluster contributions are jointly sign-symmetric, meaning $(\epsilon_1\zeta_1, \ldots, \epsilon_G\zeta_G) \overset{d}{=} (\zeta_1, \ldots, \zeta_G)$ for every $\epsilon \in \mathcal{E}$, then the sign-change test is exactly level α in finite samples, and so is its Monte Carlo version using $B - 1$ random sign vectors together with the identity. Sign symmetry holds for the infeasible statistic built from the true errors under* (3.2) *whenever the cluster error contributions are independent across g and symmetric about zero. The feasible statistic differs from the infeasible one by $O_p(G^{-1/2})$ and is asymptotically level α as $G \to \infty$.*

Exactness for the feasible version does not follow from Theorem 2, and we do not claim it. The finite-sample statement applies to the infeasible statistic; the feasible statistic inherits

validity only asymptotically in G, by the approximate-symmetry argument of Canay et al. (2017). Section 8 measures the finite-sample gap.

The second procedure is the restricted wild cluster bootstrap of Cameron et al. (2008), drawing Rademacher weights at the cluster level with (3.2) imposed on the residuals. MacKinnon and Webb (2017) document the conditions under which it fails, and those conditions concern cluster heterogeneity rather than the ratio m/G that drives Corollary 1.

7 Power, the minimum detectable response, and the optimal statistic

Let the alternative be $\gamma_\ell = \kappa\, k(\ell)$ for a response path $k(\cdot) \geq 0$ and a scalar magnitude $\kappa > 0$, so that $H_1 : \kappa > 0$. Under local alternatives the noncentrality of the Wald statistic depends on the design through two scalars: $\Sigma_\delta^2 = \sum_c (d_c - \bar{d})^2$, a property of the exposure index, and a timing quantity I_k determined by the treatment-timing design and the horizon window.

Corollary 2 (closed-form minimum detectable response)**.** *Let $\lambda^*(\alpha, \beta, m)$ solve $\Pr\{\chi^2_m(\lambda^*) > \chi^2_{m,1-\alpha}\} = 1 - \beta$. With a linear wedge and homoskedastic categories,*

$$\kappa^{\text{Wald}}_{\text{MDE}} = \sigma \sqrt{\frac{\lambda^*(\alpha, \beta, m)}{\Sigma_\delta^2\, I_k}}, \qquad \kappa^{\text{dir}}_{\text{MDE}} = \sigma\, \frac{z_{1-\alpha} + z_\beta}{\sqrt{\Sigma_\delta^2\, I_k}}. \tag{7.1}$$

Both Σ_δ^2 and I_k are computable before any outcome is observed, and σ^2 is the only quantity requiring pilot data, so (7.1) is an ex-ante design rule. The expression for $\kappa^{\text{Wald}}_{\text{MDE}}$ is stated against the chi-square critical value, which Proposition 1 shows is the wrong one; it therefore understates the minimum detectable response of any correctly calibrated Wald test. Replacing $\chi^2_{m,1-\alpha}$ by the exact critical value from (4.1) raises the required noncentrality from 17.85 to 41.46 at $G_{\text{eff}} = 36.42$ and $m = 13$, inflating the minimum detectable response by a factor of 1.524.

The Wald statistic is not the right statistic against this alternative.

Theorem 3 (optimal directed test)**.** *Suppose $V = \text{Var}(\hat{\gamma}_P)$ and the response shape k_P are known. Among tests of $H_0 : \kappa = 0$ against $H_1 : \kappa > 0$ that are invariant under shifts of the common event-study component and depend on the data only through the maximal invariant, the test rejecting when $t = k_P' V^{-1} \hat{\gamma}_P / \sqrt{k_P' V^{-1} k_P} > z_{1-\alpha}$ is uniformly most powerful, with power $\Phi(\kappa \sqrt{k_P' V^{-1} k_P} - z_{1-\alpha})$.*

When k_P is restricted only to the non-negative orthant no uniformly most powerful invariant test exists, and the likelihood-ratio statistic has the chi-bar-square null distribution of Kudô (1963) and Shapiro (1988), a mixture $\sum_{j=0}^{m} w_j(V) \chi^2_j$ in which $w_j(V)$ is the probability

that the projection of $N(0, V)$ onto the non-negative orthant in the V^{-1} metric has exactly j positive coordinates. Its power lies between the directed test and the two-sided Wald.

Calibrating each test so that it delivers size α, the ratio of minimum detectable responses between the directed and Wald tests is 0.393 at $G_{\text{eff}} = 36.42$, 0.448 at $G = 51$ and 0.516 at $G = 100$. Directing the test cuts the detectable response by roughly 61% at the smallest of these designs.

8 Simulations

Design. We simulate a two-layer panel. A latent factor $F_{it} = f_i + g_t + \tau D_{it} + v_{it}$ drives C category outcomes $Y^*_{itc} = a_{ic} + \lambda_{tc} + \theta_c F_{it} + \varepsilon_{itc}$, and the recorded outcome adds a time-invariant unit-by-category distortion and a treatment-induced term, $Y_{itc} = Y^*_{itc} + \mu_{ic} + \kappa k(\ell_{it}) d_c D_{it} + u_{itc}$. All disturbances are Gaussian and independent across clusters. The null (3.2) corresponds to $\kappa = 0$. We vary $G \in \{20, 30, 51, 100\}$ with one unit per cluster, $C \in \{3, 5, 8\}$, and set $T = 120$ with three adoption cohorts, $q = 18$ included horizons and $m = 13$ post-treatment restrictions. Exposure indices are equally spaced. Every size cell uses 2,000 replications at nominal level 0.05; permutation tests enumerate all $C!$ label assignments and the bootstrap and sign-change tests use 1,999 draws. Monte Carlo standard errors for a rejection rate near 0.05 are 0.005.

Size. Table 1 reports null rejection rates. The chi-square calibration rejects at 0.920, 0.693, 0.393 and 0.172 at $G = 20$, 30, 51 and 100 with $C = 3$, and the pattern is nearly identical across C, consistent with Proposition 1, in which C enters only through ω_2. The F calibration reduces but does not remove the distortion. The permutation test is at nominal size for $C \in \{5, 8\}$, ranging from 0.0425 to 0.059 across all eight such cells. At $C = 3$ it rejects exactly 0.000 of the time, which is Proposition 2(iii) rather than a numerical artifact. The wild cluster bootstrap ranges from 0.0505 to 0.067 and the sign-change test from 0.0215 to 0.066, the latter conservative at $G = 20$.

Accuracy of the exact law. Evaluating (4.1) at G_{eff} from Definition 1 predicts a null rejection rate of 0.378 against the observed 0.389 at $G = 51$, $C = 5$, $m = 13$. Across the twelve (G, C) cells the mean absolute prediction error is 2.37 percentage points using G_{eff} and 13.13 percentage points using G. In this design $G_{\text{eff}} = 36.42$, or 71.4% of G, and it is a design object rather than a random one: its standard deviation across replications is 10^{-13}. The exact Hotelling calibration evaluated at G_{eff} rejects at 0.0445 to 0.0470 at $G = 51$ and is conservative at $G = 20$, where it rejects at 0.000.

The design rule. Applying Corollary 1 through (4.1), the largest number of restrictions holding chi-square size below 0.075 is $m = 2$ at $G_{\text{eff}} = 36.4$, $m = 4$ at 72.7, $m = 6$ at 100 and

$m = 8$ at 150. Testing $m = 13$ restrictions at $G_{\text{eff}} = 36.4$ gives a predicted size of 0.379.

When exchangeability fails. We vary the ratio of the largest to smallest category error variance, which violates Assumption 3. At $C = 5$ the permutation test using the Wald statistic rejects at 0.050, 0.0635, 0.0772 and 0.0901 for variance ratios 1, 2, 4 and 9. The distortion is monotone in the ratio and is an order of magnitude smaller than the chi-square distortion it replaces.

The price of label randomization. Relative to a normal benchmark, the permutation test requires a noncentrality inflated by 1.892 at $C = 4$, 1.352 at $C = 5$, 1.235 at $C = 6$, 1.149 at $C = 8$ and 1.098 at $C = 10$. Exactness is bought with power, and the price falls as C grows.

Power. Table 2 reports size-adjusted power at $G = 51$ against a linear recording response of magnitude κ, over 600 replications per cell. Two patterns are visible and they point in opposite directions. Label randomization with the two-sided Wald statistic is the least powerful procedure at every κ and every C we ran: at $C = 5$ and $\kappa = 0.45$ it attains 0.383 against 0.705 for the chi-square statistic at its size-adjusted critical value. Directing the statistic more than reverses that loss. The directed cluster sign-change test attains 0.983 in the same cell, and at $C = 8$ it reaches 0.842 at $\kappa = 0.20$ where the two-sided Wald reaches 0.277. A researcher who buys exactness with label permutation and then spends it on a two-sided statistic is worse off than before; the gain comes from the direction restriction, not from the randomization.

The size-adjusted power of the two-sided permutation test is non-monotone in κ at $C = 5$, falling from 0.452 at $\kappa = 0.60$ to 0.383 at $\kappa = 0.80$. The Monte Carlo standard error at these rates is 0.020, so the decline is about three standard errors and we do not attribute it to simulation noise. We have not established its cause.

9 Application

We apply the procedures to two state panels assembled for a companion project on treatment-induced recording change. The identification claim in each case is that under measurement invariance the post-treatment event-study coefficients do not vary with the exposure index, so (3.2) holds. The application is illustrative of the inference problem rather than a contribution to either substantive literature.

The first panel covers 51 states from 1979 to 2024 with six offense categories graded by the share of victimizations reported to police, clustered on state, so $G = 51$, $C = 6$ and $m = 9$. The design has $G_{\text{eff}} = 45.9$, which by the rule of Section 8 supports $m = 3$ restrictions at size 0.075; the specification tests $m = 9$, for which (4.1) predicts a chi-square size of 0.169. The chi-square test of (3.2) rejects, with a statistic of 19.91 and $p = 0.018$. No correctly sized procedure

Table 1: Null rejection rates, nominal level 0.05

		Clusters G			
Calibration	C	20	30	51	100
Chi-square	3	0.920	0.693	0.393	0.172
	5	0.905	0.686	0.389	0.187
	8	0.917	0.690	0.356	0.172
F	3	0.838	0.586	0.311	0.143
	5	0.821	0.558	0.309	0.153
	8	0.834	0.571	0.283	0.138
Label permutation	3	0.000	0.000	0.000	0.000
	5	0.053	0.050	0.046	0.059
	8	0.051	0.051	0.043	0.058
Wild cluster bootstrap	3	0.053	0.055	0.053	0.061
	5	0.064	0.062	0.051	0.067
	8	0.061	0.061	0.051	0.056
Cluster sign-change	3	0.027	0.036	0.046	0.058
	5	0.022	0.041	0.050	0.066
	8	0.033	0.041	0.048	0.053

Note: rejection rates for the joint null $H_0 : \gamma_P = 0$ of (3.2) in the pooled gradient regression (3.1), estimated by OLS with the CR1 covariance clustered on g. Design: $m = 13$ post-treatment restrictions, $q = 18$ included horizons, $T = 120$, three adoption cohorts, one unit per cluster, equally spaced exposure indices. Label permutation randomizes the $C!$ assignments of the exposure index to category data arrays and enumerates all of them; the wild cluster bootstrap and the cluster sign-change test use 1,999 draws. 2,000 replications per cell; Monte Carlo standard error at a rate of 0.05 is 0.005. Source: `results/theory/inference/size_table.csv`.

rejects: the exact Hotelling calibration at G_{eff} returns $p = 0.097$, the label-permutation test over all 720 assignments returns $p = 0.096$, the wild cluster bootstrap returns $p = 0.075$ and the cluster sign-change test returns $p = 0.083$. The reported finding is an artifact of the reference distribution.

A pre-period placebo in the same specification behaves differently. Its chi-square p-value is 9×10^{-6} and its label-permutation p-value is 0.0167, which is the smallest attainable value at $C = 6$ under Proposition 2(ii). The placebo rejection survives correct calibration, so the pre-period contamination in that design is not an inference artifact.

The second panel covers 51 states from 2011 to 2024 with health conditions graded by whether a clinical encounter is required for the respondent to report them. Two conditions are surveyed only in odd years, so Assumption 1 fails on the full category set and the exact procedures of Sections 5 and 6 do not apply to the specification as originally run. We report two balanced sub-designs instead: five annually surveyed categories, and all seven categories on odd years only. Neither rejects (3.2) under any calibration. Both have G_{eff} near 33 and

Table 2: Size-adjusted power against the recording response κ, $G = 51$

		Recording response κ					
Statistic and calibration	C	0.05	0.10	0.15	0.20	0.30	0.45
Wald, chi-square	5	0.082	0.107	0.145	0.212	0.362	0.705
	8	0.058	0.113	0.150	0.277	0.537	0.850
Wald, wild cluster bootstrap	5	0.080	0.110	0.147	0.220	0.362	0.715
	8	0.053	0.112	0.157	0.273	0.532	0.852
Wald, label permutation	5	0.083	0.103	0.138	0.170	0.212	0.383
	8	0.055	0.097	0.142	0.187	0.395	0.655
Directed, label permutation	5	0.147	0.250	0.403	0.518	0.657	0.797
	8	0.175	0.370	0.535	0.732	0.890	0.973
Directed, cluster sign-change	5	0.143	0.320	0.527	0.688	0.895	0.983
	8	0.155	0.403	0.613	0.842	0.968	0.997

Note: size-adjusted rejection rates against the alternative $\gamma_\ell = \kappa k(\ell)$ with a linear response path, for the joint null $H_0 : \gamma_P = 0$ of (3.2). Each entry uses the critical value that delivers rejection rate 0.05 for that procedure under $\kappa = 0$ in the same design, so entries are comparable across rows. Design: $G = 51$ clusters, $m = 13$ post-treatment restrictions, $q = 18$ horizons, $T = 120$, three adoption cohorts, equally spaced exposure indices. Label permutation enumerates all $C!$ assignments; the cluster sign-change test uses 1,999 draws. 600 replications per cell; Monte Carlo standard error at a rate of 0.5 is 0.020. Source: `results/theory/inference/power_curves.csv`.

support $m = 2$ restrictions against the $m = 7$ tested.

Neither non-rejection is evidence that measurement invariance holds. The minimum detectable response from (7.1) exceeds the magnitude implied by the substantive mechanism in both panels, and we report that comparison rather than describing the tests as passed.

10 Conclusion

The cluster-robust Wald statistic for a cross-outcome falsification test is a scaled Hotelling T^2, so its chi-square calibration is valid only when the number of restrictions grows more slowly than the two-thirds power of the number of clusters. At 51 clusters and 13 restrictions the nominal 5% test rejects a true null 38.9% of the time. Three procedures restore size: an exact label-permutation test valid in finite samples under category exchangeability, a cluster sign-change test, and the restricted wild cluster bootstrap. In a crime-reporting design the chi-square test rejects at $p = 0.018$ and none of the three does.

The main limitation is Assumption 3. Exactness of the label-permutation test requires the category data arrays to be exchangeable given the design, which fails when categories have different error variances, and our simulations put the size at 0.090 when the largest category variance is nine times the smallest. The cluster-level procedures do not require it, and they are

the default when the exchangeability condition is doubtful or when $C \leq 4$ makes permutation unusable.

One extension is immediate. Theorem 1 requires a category-balanced design, and the health application shows that requirement binding in practice. Extending the exact reduction to panels unbalanced across categories would recover the procedure for designs where categories are surveyed on different schedules.

A Proofs

Proof of Lemma 1. Under Assumption 1 the regressor block in (3.1) is common across categories after the two-way within transformation, so stacking the C categories gives design matrix $[\,\mathbf{1}_C \otimes \tilde{X},\ \delta \otimes \tilde{X}\,]$ with $\delta = (\delta_1, \ldots, \delta_C)'$. Because $\mathbf{1}_C'\delta = 0$ by the centering of δ, the two blocks are orthogonal, and the normal equations separate into $(\mathbf{1}_C'\mathbf{1}_C)S\hat{\beta} = \sum_c \tilde{X}'\tilde{y}_c$ and $(\delta'\delta)S\hat{\gamma} = \sum_c \delta_c \tilde{X}'\tilde{y}_c$. Dividing by C and $\sum_c \delta_c^2$ respectively and using $b_c = S^{-1}\tilde{X}'\tilde{y}_c$ gives the stated expressions. □

Proof of Proposition 1. Write the CR1 covariance estimator as $\hat{V} = \omega_2^{-1}\,\frac{G}{G-1}\,A\big(\sum_g a_g a_g'\big)A'$ with A the relevant rows of S^{-1} and a_g the cluster score. Under Assumption 2(ii) each cluster carries design weight $S_g = S/G$, so $Aa_g = G^{-1}\eta_g$ up to the common factor absorbed in A, and the gradient estimator satisfies $\hat{\gamma}_P = \bar{\eta}_P$ in the same normalization. Substituting,

$$W = \hat{\gamma}_P'\hat{V}_{PP}^{-1}\hat{\gamma}_P = \omega_2\,\bar{\eta}_P'\Big[\tfrac{1}{G-1}\textstyle\sum_g(\eta_{gP} - \bar{\eta}_P)(\eta_{gP} - \bar{\eta}_P)'\Big]^{-1}\bar{\eta}_P\,G = \omega_2 T^2,$$

where the centering appears because the null (3.2) is imposed on the residuals. This is an algebraic identity and does not use Assumption 2(i).

For the distribution, Assumption 2(i) gives $\sqrt{G}\,\bar{\eta}_P \sim N(0, \Sigma_{PP})$ and $(G-1)\hat{S}_P \sim W_m(\Sigma_{PP}, G-1)$ independently, by Cochran's theorem applied to the i.i.d. Gaussian sample $\eta_{1P}, \ldots, \eta_{GP}$. Hence T^2 is Hotelling's T^2 with dimension m and $G-1$ degrees of freedom (Hotelling 1931), and the standard transformation $\frac{G-m}{(G-1)m}T^2 \sim F(m, G-m)$ holds for $G - m \geq 1$. Substituting $T^2 = W/\omega_2$ gives (4.1). □

Proof of Corollary 1. By (4.1) the exact size of the nominal chi-square test is $\Pr\{F(m, G-m) > (G-m)c_m/(\omega_2(G-1)m)\}$ with $c_m = \chi^2_{m,1-\alpha}$. Write $F(m, G-m) \overset{d}{=} (\chi^2_m/m)/(\chi^2_{G-m}/(G-m))$ with the two chi-squares independent, so the size equals $\Pr\{\chi^2_m > c_m\,\omega_2^{-1}(G-1)^{-1}\chi^2_{G-m}\}$.

For (i), fix m and expand $\chi^2_{G-m}/(G-1) = 1 + O_p(G^{-1/2})$ with mean $1 - (m-1)/(G-1)$. A second-order Taylor expansion of $u \mapsto \Pr\{\chi^2_m > c_m u\}$ around $u = 1$, together with $\sqrt{m}|\log\omega_2| \to 0$, gives the stated expansion, in which the $O(G^{-1})$ term carries coefficient $c_m f_m(c_m)(m + c_m)/2 > 0$.

For (ii), take logarithms. The event is $\log \chi^2_m - \log \chi^2_{G-m} > \log c_m - \log(\omega_2(G-1))$. With $m \to \infty$, $\log \chi^2_m$ is asymptotically normal with mean $\log m - 1/m + O(m^{-2})$ and variance $2/m$, and similarly for $\log \chi^2_{G-m}$ with $G - m \to \infty$. Standardizing and using $\log c_m = \log m + \sqrt{2/m}\, z_{1-\alpha} + o(m^{-1/2})$ gives size $\to \Phi(\chi - z_{1-\alpha})$ with $\chi = \lim \sqrt{m/2}\, \log\{(G-1)/(G-m)\}$. Expanding $\log\{(G-1)/(G-m)\} = (m-1)/(G-1) + O((m/G)^2)$ gives $\chi_{m,G} = m^{3/2}/(\sqrt{2}G) + o(1)$ when $m^{5/2}/G^2 \to 0$. If $m/G \to r \in (0,1)$ then $\log\{(G-1)/(G-m)\} \to -\log(1-r) > 0$ and $\chi_{m,G} \to \infty$, so the size tends to one. □

Proof of Theorem 1. The argument is the classical group-invariance construction; see Lehmann and Romano (2005, Ch. 15). Order-invariance of $\mathcal{T}$ gives, for every $\pi \in S_C$, $\mathcal{T}(\mathcal{D}, \delta_\pi) = \mathcal{T}(\mathcal{D}_{\pi^{-1}}, \delta)$ where $\mathcal{D}_{\pi^{-1}} = (\mathcal{Y}_{\pi^{-1}(1)}, \ldots, \mathcal{Y}_{\pi^{-1}(C)})$: the statistic is computed from the multiset $\{(\delta_{\pi(c)}, \mathcal{Y}_c)\}_c$, and re-indexing by $j = \pi(c)$ turns this into $\{(\delta_j, \mathcal{Y}_{\pi^{-1}(j)})\}_j$.

Under Assumption 3, $\mathcal{D}_{\pi^{-1}} \overset{d}{=} \mathcal{D}$ conditional on $\mathcal{Z}$ for each fixed π. Since S_C is a finite group acting on the index set, the vector $(T(\pi))_{\pi \in S_C}$ is exchangeable conditional on $\mathcal{Z}$: for any $\sigma \in S_C$, $(T(\sigma\pi))_\pi \overset{d}{=} (T(\pi))_\pi$. Exchangeability of the M values implies that the rank of $T(\mathrm{id})$ among them is uniform on $\{1, \ldots, M\}$ when ties have probability zero, and stochastically no larger than uniform otherwise. Rejecting when $T(\mathrm{id})$ exceeds the kth order statistic with $k = \lceil (1-\alpha)M \rceil$ therefore has conditional probability $(M-k)/M \le \alpha$, and $\lfloor \alpha M \rfloor / M$ under distinctness. Averaging over $\mathcal{Z}$ preserves the bound. The randomized version distributes the residual mass $\alpha - \lfloor \alpha M \rfloor / M$ over the ties and attains α exactly. □

Proof of Proposition 2. (i) The identity permutation contributes one count to the numerator of $\hat{p}$, so $\hat{p} \ge 1/M$ with $M = C!$.

(ii) With equally spaced d we may write $\delta_c = s\,(c - \frac{C+1}{2})$ for a scale $s > 0$, so the order-reversing permutation $r(c) = C + 1 - c$ satisfies $\delta_{r(c)} = -\delta_c$. Pre-composition negates the label vector: $(\delta_{r\circ\pi})_c = \delta_{r(\pi(c))} = -\delta_{\pi(c)}$, so $\delta_{r\circ\pi} = -\delta_\pi$. Evenness of $\mathcal{T}$ then gives $T(r \circ \pi) = T(\pi)$ for every π. The map $\pi \mapsto r \circ \pi$ is left translation by an involution without fixed points, hence a fixed-point-free involution on S_C, so the M values fall into $M/2$ equal pairs and every attainable $\hat{p}$ is a multiple of $2/M$.

For the Wald statistic, $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ by Lemma 1, each cluster score is odd in δ, and the CR1 covariance is quadratic in the cluster scores, so $\hat{V}_\gamma(-\delta) = \hat{V}_\gamma(\delta)$ and the quadratic form is unchanged.

(iii) Rejection at level α requires some attainable $\hat{p} \le \alpha$, hence $2/C! \le \alpha$ in the even case and $1/C! \le \alpha$ otherwise. □

Proof of Theorem 2. For the first claim, replace S_C in the proof of Theorem 1 by the finite group $\mathcal{E} = \{-1, +1\}^G$ acting by coordinatewise sign change. That proof uses only finiteness of the group and invariance of the data law under it, both of which the sign-symmetry hypothesis supplies.

For the second claim, let u^0 denote the true errors under (3.2). The infeasible contribution ζ_g^0 is a fixed linear functional of cluster g's errors alone, so independence across g together with symmetry of each summand about zero gives $(\epsilon_g \zeta_g^0)_g \stackrel{d}{=} (\zeta_g^0)_g$ for every $\epsilon \in \mathcal{E}$.

For the third claim, the feasible contribution replaces u^0 by the restricted residual, which subtracts a term $S^{-1} S_g[\sum_h (\cdot)_h]$. The factor $\|S_g\|/\|S\|$ is $O(G^{-1})$ and it multiplies a sum of G cluster contributions, which is $O_p(G^{1/2})$ times a single one; relative to ζ_g itself the discrepancy is therefore $O_p(G^{-1/2})$. Asymptotic level α follows from the approximate-symmetry theorem of Canay et al. (2017), whose conditions require the randomization distribution to converge to one that is invariant under $\mathcal{E}$, which the $O_p(G^{-1/2})$ bound delivers. □

Proof of Corollary 2. Under the stated alternative and homoskedasticity, $\hat{\gamma}_P$ is asymptotically normal with mean κk_P and covariance $V = \sigma^2(\Sigma_\delta^2 I_k)^{-1}\Omega_P$ for a design matrix Ω_P normalized so that the linear wedge gives $k_P' \Omega_P^{-1} k_P = 1$. The Wald statistic is then noncentral chi-square with m degrees of freedom and noncentrality $\lambda = \kappa^2 \Sigma_\delta^2 I_k / \sigma^2$. Setting $\lambda = \lambda^*(\alpha, \beta, m)$ and solving for κ gives the first expression in (7.1). For the directed statistic, t is standard normal under (3.2) and normal with mean $\kappa\sqrt{\Sigma_\delta^2 I_k}/\sigma$ under the alternative, so power $1-\beta$ at level α requires that mean to equal $z_{1-\alpha} + z_\beta$, which gives the second expression. □

Proof of Theorem 3. Invariance under shifts of the common event-study component reduces the data to the maximal invariant $\hat{\gamma}_P$, which is $N(\kappa k_P, V)$ with V and k_P known. The family in κ has density proportional to $\exp\{\kappa k_P' V^{-1} \hat{\gamma}_P - \frac{1}{2}\kappa^2 k_P' V^{-1} k_P\}$ relative to the $\kappa = 0$ density, a one-parameter exponential family with natural sufficient statistic $k_P' V^{-1} \hat{\gamma}_P$ and natural parameter κ. By the Neyman-Pearson lemma the most powerful test against any fixed $\kappa > 0$ rejects for large values of that statistic, and the rejection region does not depend on which $\kappa > 0$ is chosen, so the test is uniformly most powerful among invariant tests. Standardizing by $\sqrt{k_P' V^{-1} k_P}$ gives $t \sim N(\kappa\sqrt{k_P' V^{-1} k_P}, 1)$, from which the size and the stated power function follow. □

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